

Machine Learning Final Assessment e-Portfolio Report

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Module: Machine Learning

Submission: Final Assessment / e-Portfolio

e-Portfolio GitHub Link: [GitHub e-Portfolio Branch](#)

Word count note: the reflective section is approximately 780 words and is kept within the 1,000-word cap.

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1. Introduction

This e-portfolio report presents evidence of my learning and development across the Machine Learning module. It is linked to the GitHub e-portfolio branch shown on the cover page, where the supporting notebooks, reports, presentation material and reflective artefacts are organised by unit. The portfolio demonstrates progression from exploratory data analysis and classical statistical learning into clustering, neural networks, convolutional neural networks and model performance evaluation. It also demonstrates professional learning: effective machine learning practice depends not only on executing algorithms, but also on understanding data limitations, evaluating models critically, communicating findings clearly and considering legal, social, ethical and professional implications.

2. Artefact and Evidence Overview

The artefacts are organised by unit to show progressive development. The table below summarises the main evidence included or to be included in the GitHub e-portfolio. Where forum participation was not possible, I have included a retrospective artefact so that the learning outcome is still addressed honestly and transparently.

Table 1. Unit-by-unit evidence map for the e-portfolio.

Unit	Portfolio evidence	Attachment / location	Primary learning evidence
1	Retrospective initial post on Industry 4.0/5.0 and information system failure	Appendix A; GitHub reflective post	Ethical, professional and social issues
2	Auto-mpg EDA: missing values, skewness/kurtosis, heatmap, scatter plots and categorical encoding	Unit02 notebook and datasets	Dataset applicability and EDA
3	Covariance, Pearson correlation, simple, multiple and polynomial regression notebooks	Unit03 Ex1-Ex4 notebooks	Regression and correlation foundations
4	Fuel consumption demo and Global Population/GDP correlation-regression task	Unit04 notebook and CSV files	Regression using real-world datasets
5	Jaccard coefficient calculation	Appendix B table	Similarity measurement
6	K-means clustering and Airbnb analytical report	Analytical report; Airbnb notebook/report	Unsupervised learning and business interpretation
7	Simple perceptron, AND perceptron and multi-layer perceptron activities	Unit07 Ex1-Ex3 notebooks	Neural network foundations
8	Gradient descent cost-function notebook and ANN/insurance ethics reflection	Unit08 notebook; reflective notes	Optimisation and AI ethics
9	CNN object recognition activity and ethical/social reflection	CIFAR-10 notebook/PDF	CNN application and responsible use
10	CNN layer/image recognition continuation	CIFAR-10 evidence and presentation material	Feature learning and data distribution
11	Model performance measurement and final CIFAR-10 CNN project	Unit11 notebook; presentation; transcript	Performance evaluation and critical appraisal
12	Future of ML reflection: self-supervised learning, NAS, Edge AI, ethics and sustainability	Final reflection section	Future-facing professional awareness

Table 2. Summary of major individual project contributions.

Project artefact	Focus	Contribution evidence	Learning value
Airbnb analytical report	Structured tabular data;	EDA, K-means clustering,	Showed how ML can support

Project artefact	Focus	Contribution evidence	Learning value
	business segmentation	dataset limitation discussion, strategic interpretation	pricing, host engagement and policy decisions while avoiding overclaiming due to missing revenue/booking data
CIFAR-10 final project	Image data; object recognition	Baseline CNN, improved CNN, augmentation, batch normalisation, dropout, callbacks, precision/recall/F1 and confusion matrix	Showed stronger model evaluation maturity and understanding of overfitting, generalisation and class-level weaknesses

3. Individual Contributions

My main individual contributions were the Airbnb analytical report and the CIFAR-10 CNN object recognition project. In the Airbnb work, I framed a business question around whether New York City listings could be segmented using price, availability, review activity and host scale. I treated review variables as imperfect demand proxies because the dataset did not include bookings, occupancy or realised revenue. This was important because it prevented overclaiming and kept the analysis aligned to the evidence available. I used exploratory analysis and clustering to translate listing behaviour into strategic implications such as pricing guidance, host support and policy monitoring.

In the final CNN project, I implemented an object recognition workflow using the CIFAR-10 dataset. I normalised pixel values, flattened class labels and created a stratified training-validation split to preserve class balance. I first built a baseline CNN to establish a benchmark. The baseline achieved 74.26% test accuracy, but the training and validation curves showed overfitting. I then built an improved CNN using data augmentation, batch normalisation, dropout, early stopping and learning-rate scheduling. The improved model achieved 82.51% test accuracy and a macro F1-score of 82.26%, showing stronger generalisation. I also interpreted the confusion matrix and class-level metrics, identifying stronger performance on visually distinctive classes and weaker performance on visually similar animal classes.

Table 3. Final project model comparison: baseline CNN vs improved CNN.

Metric	Baseline CNN	Improved CNN	Change
Final training accuracy	88.74%	80.87%	-7.87 pp
Final validation accuracy	73.72%	82.33%	+8.61 pp
Best validation accuracy	74.25%	83.01%	+8.76 pp
Test accuracy	74.26%	82.51%	+8.25 pp
Final validation loss	0.9557	0.5075	-0.4482
Test loss	0.9621	0.5152	-0.4469

4. Evaluation of Earlier Project Work vs Final Project

The earlier analytical work and the final CNN project represent two different stages of learning. The Airbnb task used structured tabular data and focused on business segmentation. K-means was appropriate because the aim was to group listings into meaningful operational categories. However, the output depended on feature selection, scaling and the chosen number of clusters. The main risk was interpretability: clusters can look precise even when the underlying dataset lacks direct commercial outcomes.

The final CIFAR-10 project involved unstructured image data and required a different modelling approach. A CNN was more suitable because it can learn spatial image features directly from pixels. The final project

also required a more mature evaluation strategy. Instead of relying on a single score, I compared learning curves, test accuracy, test loss, precision, recall, F1-score and the confusion matrix. This demonstrated stronger critical appraisal of technical risk and uncertainty. The progression from K-means segmentation to CNN evaluation shows that my understanding moved from running machine learning methods to selecting, validating and explaining them more critically.

Table 4. Evaluation of earlier project work compared with the final CNN project.

Dimension	Earlier project / Unit 6	Final project / Unit 11
Dataset type	Structured tabular listing data	Unstructured image data
Main method	K-means clustering and business interpretation	Convolutional neural network classification
Primary risk	Cluster interpretability and incomplete commercial outcomes	Overfitting, generalisation and class-level errors
Evaluation emphasis	Business usefulness and segmentation logic	Learning curves, test performance, F1-score and confusion matrix
Development shown	Understanding of unsupervised learning and data limitations	Stronger technical evaluation and critical appraisal

5. Reflective Section

What? This module developed my understanding of machine learning through a sequence of practical activities. I began with exploratory data analysis, correlation and regression, then moved into clustering, similarity measures, perceptrons, gradient descent, artificial neural networks, convolutional neural networks and model performance evaluation. The two strongest artefacts in my portfolio are the Airbnb analytical report and the CIFAR-10 CNN object recognition project. The Airbnb report helped me understand how machine learning can support business strategy when data is structured and interpretable. The CIFAR-10 project helped me understand the additional complexity involved in deep learning, especially validation design, overfitting control and class-level evaluation. Across the module, I also reflected on ethical and professional issues such as fairness, privacy, transparency, resilience and the responsible deployment of AI in domains such as insurance and image recognition.

So what? The most important technical lesson was that model selection must be driven by the dataset and the problem. Regression and clustering were suitable for structured Airbnb listing data because the features were tabular and the objective was business insight. CNNs were more suitable for CIFAR-10 because the task involved spatial image patterns that classical models would struggle to capture without manual feature engineering. I also learned that performance must be interpreted critically. In the CNN project, the baseline model achieved high training accuracy but showed a widening gap between training and validation performance. This changed how I define model success. A strong model is not the one that performs best on training data, but the one that generalises reliably to unseen data. The improved CNN demonstrated this clearly because its training accuracy was lower than the baseline, but its validation accuracy, test accuracy and test loss were substantially better.

A second learning point was that data limitations shape the validity of any conclusion. In the Airbnb report, the dataset was useful but incomplete because it lacked direct booking, occupancy and revenue information. This meant that review activity could only be treated as a proxy for demand. This was an important professional lesson because real-world machine learning often involves imperfect datasets, and analysts must avoid overstating what the data can prove. It also made me more aware of the ethical risk of presenting model outputs as objective truth when they are actually dependent on assumptions, missing variables and data collection choices.

Emotionally, the module was demanding because I had to balance academic work with full-time professional responsibilities. There were moments where the workload felt heavy, especially when moving from simpler models into neural networks and CNN evaluation. However, each completed notebook and report increased

my confidence because the practical outputs made the concepts more concrete. This was my second attempt at the module, and it became a strong learning experience in work ethic, planning and prioritisation. In my first attempt, I did not prioritise the master's workload early enough, whereas this time I communicated with my work supervisor earlier and created the space needed to complete the academic work more responsibly.

Although the module includes collaborative learning outcomes, most of my assessed work was completed individually. I therefore had to take ownership of planning, implementation, review and improvement. I still approached the work with a development-team mindset by documenting evidence clearly, responding to available feedback and preparing artefacts that another reviewer could understand and reproduce. This changed how I view professional machine learning work: the value is not only in the final model, but also in whether the evidence trail is clear enough for review, challenge and future improvement.

Now what? Going forward, I will apply the module learning in both academic and professional contexts. Technically, I will pay closer attention to validation design, performance metrics, dataset limitations and ethical implications before recommending a model. I will also use clearer experiment tracking so that model changes, assumptions and results can be reviewed systematically. Professionally, the module has reinforced the importance of responsible AI adoption. As a technology lead, I can apply these lessons when evaluating AI use cases in anomaly detection, customer journeys, insurance analytics or image-based assessment. Rather than focusing only on whether a model can be built, I will consider whether the data is suitable, whether the output can be trusted, whether the model is explainable enough and what human oversight is required.

For future modules, I will maintain the e-portfolio progressively rather than treating it as an end-of-module activity. I will also capture screenshots, notebook outputs, meeting notes and reflections closer to the time the work is completed. Overall, the module helped me move from using machine learning techniques as isolated exercises to understanding them as part of a broader professional process involving data quality, algorithm selection, evaluation, ethics, communication and continuous improvement.

6. Conclusion

The e-portfolio demonstrates development across the module from foundational data analysis to applied deep learning. It shows practical engagement with EDA, regression, clustering, similarity measures, perceptrons, gradient descent, CNNs and model evaluation. The strongest development is the progression from the Airbnb analytical report to the CIFAR-10 CNN project, where I moved from business-focused segmentation to a more technically complex deep learning evaluation. The module has improved my ability to select appropriate techniques, interpret performance critically and communicate findings in a structured and responsible way.

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Appendices

The appendices provide supporting evidence and tables for the e-portfolio. They are included to improve traceability and should be treated as supporting material rather than the reflective word-count section.

Appendix A: Retrospective Unit 1 Initial Post

Industry 4.0 has significantly changed financial services and insurance by increasing the use of automation, data analytics, artificial intelligence, cloud platforms and interconnected digital channels. In insurance, these technologies support faster quotation journeys, fraud detection, claims automation, personalised pricing and improved operational monitoring. Industry 5.0 extends this discussion by emphasising human-centred, resilient and sustainable digital transformation (European Commission, 2021). This is important because financial and insurance services are not only technology-driven; they also depend on trust, fairness, accountability and customer confidence.

A relevant example of information system failure is the 2023 disruption affecting DBS Bank's digital banking services in Singapore. MAS reported that customers experienced difficulty logging in to DBS digital banking services in March 2023, and later noted that DBS experienced several major disruptions during that year (Monetary Authority of Singapore, 2023a; 2023b). The incident shows that digital transformation increases expectations for continuous availability. When systems fail, the impact is not only technical. Customers may lose access to payments, transfers and essential financial services. Organisations may also face reputational damage, regulatory scrutiny and operational cost.

For insurance, the lesson is that Industry 4.0 and 5.0 adoption must be supported by resilience engineering. AI-enabled services, digital claims platforms and customer portals should be designed with observability, failover capability, incident response processes and clear accountability. As machine learning professionals, we should not only focus on algorithmic performance. We must also consider whether the wider information system is reliable, secure, ethical and aligned with customer needs.

Appendix B: Unit 5 Jaccard Coefficient Calculation

The table below uses a simple categorical matching assumption: each matching attribute is counted as an intersection and all seven attributes are included in the union. If the tutor expects asymmetric binary Jaccard treatment, this table should be recalculated using that convention.

Table B1. Jaccard coefficient calculation under simple categorical matching.

Pair	Matching attributes	Total attributes	Jaccard coefficient
Jack, Mary	3	7	0.429
Jack, Jim	5	7	0.714
Jim, Mary	1	7	0.143

Appendix C: CNN Class-Level Performance Highlights

Table C1. Strongest and weakest CIFAR-10 classes from the improved CNN.

Category	Class	F1-score	Interpretation
Strongest	Automobile	92.34%	High visual distinctiveness
Strongest	Ship	89.54%	High visual distinctiveness
Strongest	Truck	87.42%	Distinct vehicle class
Strongest	Horse	86.51%	Clearer animal structure than cat/dog/bird
Weakest	Cat	68.88%	Often confused with dog/frog in low-resolution images
Weakest	Dog	75.53%	Confusion with cat and other

Category	Class	F1-score	Interpretation
			animals
Weakest	Bird	76.97%	Low-resolution visual ambiguity

Appendix D: Recommended Attachment Register for GitHub e-Portfolio

Table D1. Attachment register and recommended GitHub placement.

Area	Attachment / artefact	Recommended action
Final report	Machine_Learning_Final_Assessment_ePortfolio_Report.docx	Submit to portal and include in GitHub root or /final-assessment
Unit 2	Unit02 A Tutorial on Exploratory Data Analysis (1).ipynb; Unit02 auto-mpg (1).csv; Unit02 train/test CSVs	Upload under /unit02-eda
Unit 3	Unit03 Ex1-Ex4 notebooks	Upload under /unit03-regression-correlation
Unit 4	Unit04 demo notebook; FuelConsumption.csv; Global_GDP.csv; Global_Population.csv	Upload under /unit04-linear-regression
Unit 5	Jaccard calculation table	Include as markdown/notebook under /unit05-similarity
Unit 6	Analytical Report(1).docx; Airbnb_Business_Report.docx; AirbnbAnalysis.ipynb	Upload under /unit06-kmeans-airbnb
Unit 7	Unit07 Ex1 simple_perceptron; Ex2 perceptron_AND_operator; Ex3 multi-layer Perceptron	Upload under /unit07-perceptron-mlp
Unit 8	Unit08 Ex4 gradient_descent_cost_function.ipynb	Upload under /unit08-gradient-descent
Unit 9/10/11	CIFAR10_Track2_CNN_Object_Recognition.ipynb ; Colab PDF; Deep Learning presentation PDF; transcript DOCX; Unit11_model_Performance_Measurement notebook	Upload under /unit09-11-cnn-performance
Unit 12	Future of ML reflective page	Create markdown page under /unit12-future-ml